

Software Requirements Specification

HumanEval Demo Platform — Authentication, Billing & Content Services

Document Version 1.0 | Project: TestMate Demo Platform

1. Introduction

This document specifies the functional requirements for the HumanEval Demo Platform's core service modules: user authentication, billing engine, and content analysis utilities.

2. Authentication Module Requirements

REQ-AUTH-001: The system shall validate passwords against a defined security policy before acceptance.

REQ-AUTH-002: A valid password shall contain a minimum length of 8 characters.

REQ-AUTH-003: The system shall require at least one uppercase letter (A-Z) in all passwords.

REQ-AUTH-004: The system shall require at least one lowercase letter (a-z) in all passwords.

REQ-AUTH-005: The system shall require at least one numeric digit (0-9) in all passwords.

REQ-AUTH-006: Passwords shall include at least one special character from the set: ! @ # \$ % ^ & *

REQ-AUTH-007: When password validation fails, the system shall return a list of all violated rules.

REQ-AUTH-008: The system shall classify accounts as low, medium, high, or critical risk based on failed login attempts and inactivity.

REQ-AUTH-009: An account with 10 or more failed login attempts shall be classified as critical risk.

REQ-AUTH-010: An account inactive for 365 or more days shall be classified as critical risk.

REQ-AUTH-011: An account with 5 or more failed attempts, or inactive 180 or more days, shall be classified as high risk.

REQ-AUTH-012: An account with 3 or more failed attempts, or inactive 90 or more days, shall be classified as medium risk.

3. Billing Engine Requirements

REQ-BILL-001: The system shall compute order totals by applying tiered volume discounts to the unit price and quantity.

REQ-BILL-002: Orders of 100 or more units shall receive a 20 percent discount.

REQ-BILL-003: Orders of 50 or more units shall receive a 15 percent discount.

REQ-BILL-004: Orders of 20 or more units shall receive a 10 percent discount.

REQ-BILL-005: Orders of 10 or more units shall receive a 5 percent discount.

REQ-BILL-006: Orders of fewer than 10 units shall receive no discount.

REQ-BILL-007: The system shall raise a ValueError when unit_price is negative or quantity is less than one.

REQ-BILL-008: The system shall compute revenue statistics including minimum, maximum, mean, range, and median daily totals.

REQ-BILL-009: The revenue statistics function shall raise ValueError when provided an empty list.

4. Content Analysis Requirements

REQ-CONT-001: The system shall analyze text and return word count, sentence count, vowel count, and consonant count.

REQ-CONT-002: Word count shall be determined by the number of whitespace-separated tokens.

REQ-CONT-003: Sentence count shall be computed by counting occurrences of period, exclamation mark, and question mark.

REQ-CONT-004: An empty or whitespace-only input shall return zero for all metrics.

REQ-CONT-005: The system shall estimate readability grade level using average character length per word.

REQ-CONT-006: Text with average word length of 7 or more characters shall be classified as advanced.

REQ-CONT-007: Text with average word length of 5 or more characters shall be classified as intermediate.

REQ-CONT-008: Text with average word length of 3 or more characters shall be classified as basic.

REQ-CONT-009: Text with average word length below 3 characters shall be classified as elementary.

5. General Platform Requirements

REQ-GEN-001: All modules shall be testable via pytest with 100 percent function coverage.

REQ-GEN-002: The system shall provide clear error messages for all invalid input cases.

REQ-GEN-003: All floating-point return values shall be rounded to at most four decimal places.